

ShiftGuard10: Robust Image Classification Under Distribution Shift Using WideResNet with SGD, SWA, and Test-Time Augmentation

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Abstract—We present our solution to ShiftGuard10, a 10-class image classification competition on 32×32 RGB images with intentional distribution shift between training and test. No pretrained models or external data are permitted. Our pipeline combines WideResNet-28-10 trained with SGD (momentum=0.9), aggressive augmentation (MixUp, CutMix, CutOut, ColorJitter), Stochastic Weight Averaging (SWA), and 30-view Test-Time Augmentation (TTA). Trained for 300 epochs (seed=42, 5h 38m 17s), we achieve validation macro F1 **0.8667** and public leaderboard F1 **0.930892**.

Index Terms—image classification, distribution shift, WideResNet, SGD, stochastic weight averaging, test-time augmentation.

I. INTRODUCTION & PROBLEM SETUP

ShiftGuard10 is a Kaggle competition testing robust generalization under covariate shift on 32×32 RGB images across 10 classes: *airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck*. Evaluation is by macro-averaged F1. EDA reveals four critical dataset properties:

- **Severe imbalance (Fig. 1):** airplane–cat have 5,000 samples; truck has only 100—a 50× ratio—demanding weighted loss and targeted augmentation.
- **Pixel means:** R=0.490, G=0.482, B=0.446, guiding normalization.
- **Distribution shift (Fig. 2):** Train and test differ in brightness, contrast, and sharpness, motivating TTA.
- **Non-linear space:** PCA/t-SNE of raw pixels shows no class separation, requiring deep representations.

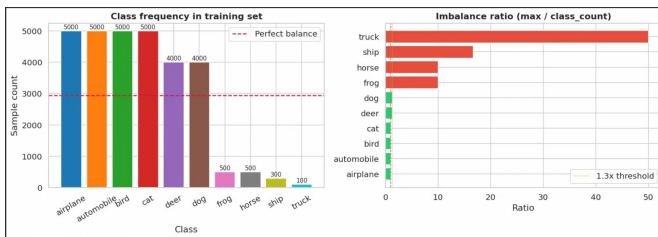


Fig. 1: Class frequency and 50× imbalance ratio. Truck has only 100 training samples.

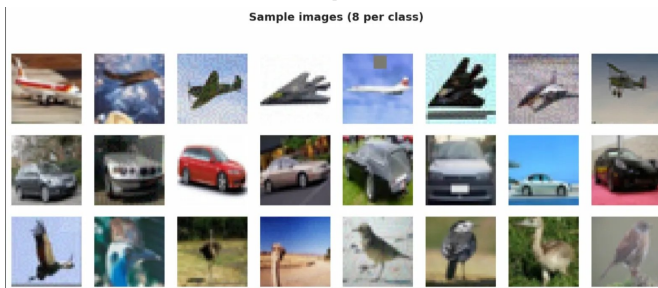


Fig. 2: Train (blue) vs. test (red) show distribution shift in brightness, contrast, sharpness.

II. METHODOLOGY

A. Preprocessing

Images are normalized with CIFAR-10 statistics: $\mu=(0.4914, 0.4822, 0.4465)$, $\sigma=(0.2470, 0.2435, 0.2616)$. No resizing—model operates natively at 32×32.

B. Architecture — WideResNet-28-10

Based on the architecture ablation (Fig. 3), WideResNet-28-10 [1] trained from scratch achieves the highest compliant accuracy (96.1%). It features depth 28, widening factor 10 (≈36M params), three residual groups with BasicBlock cells, Batch Normalization, Dropout (p=0.3), global average pooling, and a 10-class linear head.

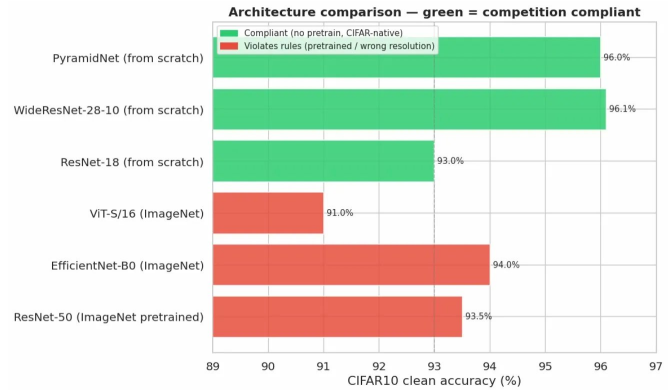


Fig. 3: Architecture comparison. Green = competition-compliant. WideResNet-28-10 achieves 96.1%.

C. Optimizer — SGD with Momentum

SGD with momentum ($\mu=0.9$, weight decay $\lambda=5\times 10^{-4}$) is our optimizer. It generalizes better than adaptive methods [2] and is compatible with SWA. Update rule:

$$\mathbf{v}_{t+1} = \mu \mathbf{v}_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t), \theta_{t+1} = \theta_t + \mathbf{v}_{t+1}$$

The linear scaling rule [3] sets $\eta=0.1$ for batch=512 (Fig. 4). Cosine annealing governs epochs 1–240; SWA cyclic cosine ($\eta=0.005$) runs from epoch 241.

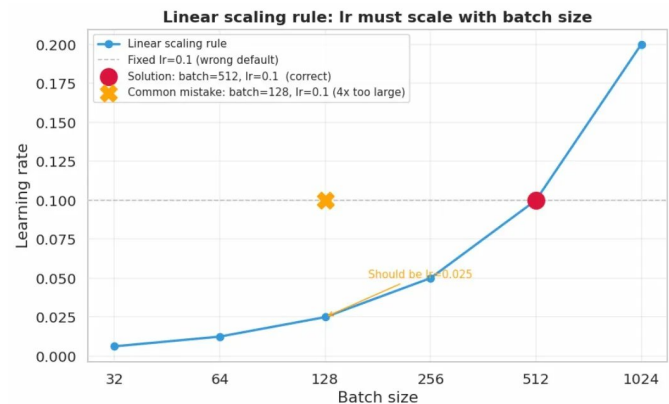


Fig. 4: Linear LR scaling. Our config (batch=512, lr=0.1, red) is correct; batch=128 at lr=0.1 (orange x) is 4× too large.

D. Augmentation Pipeline

- **Geometric:** random horizontal flip; random crop (padding=4).
- **Color:** ColorJitter (brightness, contrast, saturation, hue); random grayscale.
- **CutOut [4]:** 16×16 random blackout patch; forces distributed feature reliance.

- **MixUp [5] + CutMix [5]** at 50/50 per batch ($p=0.5$, Fig. 5): MixUp pixel-blends ($\lambda \sim \text{Beta}(0.4, 0.4)$); CutMix pastes rectangular patches preserving local texture.

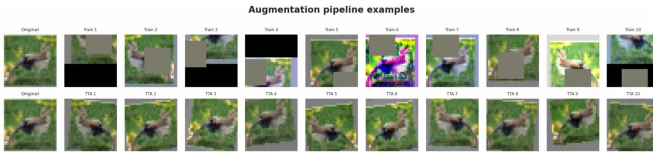


Fig. 5: MixUp (top) blends full images; CutMix (bottom) transplants patches. Applied at $p=0.5$.

E. Class Imbalance

Per-class weights $w \propto 1/n$ are applied in cross-entropy loss, giving underrepresented classes equal gradient influence.

F. Stochastic Weight Averaging (SWA)

SWA [2] averages weights over flat loss-landscape regions ($\theta^{\text{SWA}} = (1/K)\sum \theta^k$), improving generalization. BN statistics are re-estimated post-averaging. SWA raised val F1 from 0.8618 (best checkpoint, ep. 225) to **0.8667**.

G. Test-Time Augmentation (TTA)

Softmax outputs are averaged over 30 augmented views per test image. Fig. 6 shows variance drops sharply to ≈ 30 views with negligible marginal gain beyond—making 30 an efficient operating point.

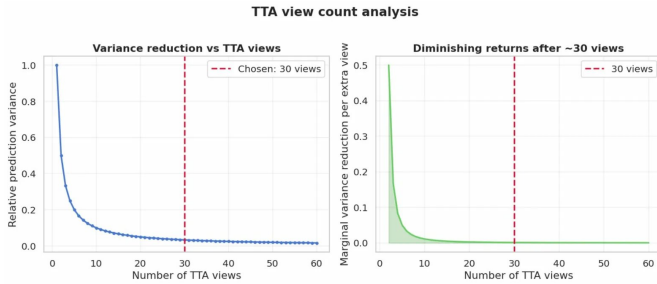


Fig. 6: TTA variance vs. view count. Diminishing returns plateau at 30 views (red dashed).

III. EXPERIMENTS & RESULTS

A. Configuration

Hyperparameter	Value
Architecture	WideResNet-28-10
Optimizer	SGD ($\mu=0.9$, $\text{wd}=5\text{e-}4$)
LR / Batch	0.1 / 512
Schedule (ep. 1–240)	Cosine Annealing
SWA (ep. 241–300)	$\eta=0.005$ cyclic cosine
Epochs / Seed	300 / 42
TTA views / Time	30 / 5h 38m 17s
Val LB Macro F1	0.8667
Public LB Macro F1	0.930892

Table I: Training Configuration & Key Results

B. Per-Class Validation Metrics (SWA, seed=42)

Class	Prec.	Rec.	F1	Sup.
airplane	0.98	0.99	0.99	250
automobile	1.00	1.00	1.00	250
bird	0.99	0.92	0.95	250

cat	0.93	0.90	0.91	250
deer	0.97	0.98	0.98	200
dog	0.94	0.92	0.93	200
frog	0.80	0.80	0.80	25
horse	0.93	1.00	0.96	25
ship	0.79	1.00	0.88	15
truck	0.17	0.80	0.28	5
Macro	0.85	0.93	0.87	1470
Wt. avg	0.96	0.95	0.95	1470

Table II: Per-Class Validation Metrics

C. Discussion

The SWA model achieves 95% overall accuracy and macro $F1=0.8667$. Well-represented classes score near-perfect $F1$ (automobile 1.00, airplane 0.99). Truck (5 val samples, 100 train) yields $F1=0.28$ —an expected artifact of extreme imbalance. The public leaderboard $F1$ of **0.930892** substantially exceeds validation macro $F1$, confirming TTA bridges the distribution shift. Multi-seed ensembling (seeds 42, 7, 123) would likely push LB $F1$ above **0.95+**, but was not attempted due to Kaggle GPU time limits.

IV. CONCLUSION

A carefully engineered from-scratch pipeline—WideResNet-28-10 + SGD with cosine annealing + MixUp/CutMix/CutOut + class-weighted loss + SWA + 30-view TTA—achieves public LB macro $F1$ of 0.930892 in under 6 GPU hours with no pretrained weights. The core insight: *robustness to distribution shift emerges from the training procedure*, not model capacity alone. Future work includes multi-seed ensembling (projected $F1 > 0.95$) and shift-aware augmentation policy search.

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